WORKED SOLUTION

STEP III 2004, Question 13

Probability and Statistics • Geometric Distribution

STEP III 2004, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q13

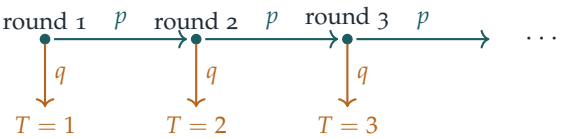
The key idea

Forget the competition for a moment and follow one competitor. The round in which he drops out is a geometric random variable, and the three competitors' drop-out rounds are independent. Every part is then a question about three independent geometric variables. The grand prize turns out to depend only on whether the *latest* drop-out round is shared, so the final part is the complement of the first two parts put together.

Solution

Write $q = 1 - p$, and assume $0 < p < 1$. For competitor j ($j = 1, 2, 3$) let T_j be the round in which he fails. He fails round k exactly when he gets through rounds $1, \ldots, k-1$ and then fails round k, so

$$\mathrm{P}(T_j = k) = p^{k-1}q, \qquad k = 1, 2, 3, \ldots$$



Since each round is passed or failed independently, and the competitors act independently, T_1, T_2, T_3 are independent. Note also that

$$\mathrm{P}(T_j \leq r - 1) = 1 - \mathrm{P}(\text{passes rounds } 1, \ldots, r-1) = 1 - p^{r-1}.$$

Part (i)

All three drop out in round k with probability $(p^{k-1}q)^3$, by independence. Summing the geometric series over k (common ratio $p^3 < 1$),

$$\mathrm{P}(T_1 = T_2 = T_3) = \sum_{k=1}^{\infty} q^3 p^{3(k-1)} = \frac{q^3}{1-p^3} = \frac{(1-p)^2}{1+p+p^2}.$$

Part (ii)

There are 3 choices for the competitor who drops out early. For a given choice, the other two both fail in round r with probability $(p^{r-1}q)^2$, and the early one fails in some round before r with probability $1 - p^{r-1}$. These events are disjoint for different choices, so

$$\mathrm{P}(\text{two in round } r,\ \text{third earlier}) = 3p^{2r-2}q^2\left(1-p^{r-1}\right).$$

Part (iii)

? **Ask yourself** Exactly when is the grand prize awarded, in terms of T_1, T_2, T_3?

Suppose the largest of the T_j is achieved by just one competitor, say $T_3 > T_1, T_2$, and let $m = \max(T_1, T_2)$. At round m competitor 3 is still in and gets through. Every other competitor still in at that round has $T_j = m$, so fails it. So competitor 3 wins the prize at round m.

Conversely, suppose the largest value is shared, say $T_2 = T_3 = M \geq T_1$. Before round M, both 2 and 3 are still in and both get through each round, so no round has a single survivor. At round M everyone remaining fails, so there is no survivor at all. So the prize is not awarded.

Hence the prize is *not* awarded exactly when the top two drop-out rounds are equal: either all three are equal (part (i)), or two are equal at some round r and the third is earlier. The second case needs $r \geq 2$, and it is the event of part (ii). Summing part (ii) over $r \geq 2$ and writing $s = r - 1$,

$$\begin{aligned}\sum_{r=2}^{\infty} 3q^2p^{2r-2}\left(1-p^{r-1}\right) &= 3q^2\sum_{s=1}^{\infty}\left(p^{2s}-p^{3s}\right) = 3q^2\left(\frac{p^2}{1-p^2}-\frac{p^3}{1-p^3}\right)\\ &= 3q^2\,\frac{p^2(1-p^3)-p^3(1-p^2)}{(1-p^2)(1-p^3)} = 3q^2\,\frac{p^2(1-p)}{(1-p^2)(1-p^3)}\\ &= \frac{3p^2q^2}{(1+p)(1-p^3)}.\end{aligned}$$

Adding part (i) and using $1 - p^3 = q(1 + p + p^2)$,

$$\mathrm{P}(\text{no prize}) = \frac{q^3(1+p)+3p^2q^2}{(1+p)(1-p^3)} = \frac{q^2(1+2p^2)}{(1+p)(1-p^3)} = \frac{(1-p)(1+2p^2)}{(1+p)(1+p+p^2)}.$$

Since $(1+p)(1+p+p^2) = 1+2p+2p^2+p^3$ and $(1-p)(1+2p^2) = 1-p+2p^2-2p^3$, their difference is $3p+3p^3$. Therefore

$$\mathrm{P}(\text{grand prize awarded}) = \frac{3p(1+p^2)}{(1+p)(1+p+p^2)}.$$

Checks

- As $p \to 0$ everyone fails round 1, and the answer to (iii) tends to 0. As $p \to 1$ ties become unlikely, and the answer tends to $\frac{3\cdot 2}{2\cdot 3} = 1$.
- Directly, the prize is awarded when one T_j beats the other two, so its probability is $3\sum_k p^{k-1}q\,(1-p^{k-1})^2$. Exact summation with $p = \frac{1}{2}$ gives $\frac{5}{7} \approx 0.714$, matching $\frac{3\cdot\frac{1}{2}\cdot\frac{5}{4}}{\frac{3}{2}\cdot\frac{7}{4}} = \frac{5}{7}$.
- A simulation of the competition itself (with 2×10^5 runs at $p = 0.6$) gave the prize in 77.9% of runs, against the formula's 0.7806.

Common slips

- Forgetting the factor 3 in part (ii) for the choice of which competitor drops out early.
- Thinking the prize needs the last two to drop out in different rounds. If two drop out together in round m and the third gets through it, he still wins. Only a tie at the *top* prevents the prize.

Beyond STEP

The question is about the maximum of independent geometric random variables and whether it is unique. For n competitors, the probability of a tie at the top does not settle down to a limit as $n \to \infty$. It oscillates very slightly with $\log n$, a well-known surprise in the analysis of algorithms such as leader election. The tools here, conditioning on one variable and summing geometric series, are developed in Grimmett and Welsh, *Probability: An Introduction*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.